Chapter 6 — Dispersion, shape, and grouping

A mean without spread is ambiguous

Learning objectives

- Compute variance and standard deviation
- Summarize by groups

Variance and standard deviation

- Variance averages squared deviations from the mean
- Example 6.18 sketches variance for a small sample; Example 6.19 takes the square root
- Larger standard deviation means more spread around the center

Example 6.18 — Variance sketch

- Example 6.18 — hands-on module
- Example 6.18 uses the sample values two, four, four, six, and eight
- Explore the chapter example module
- View files: `modules/chapter6/example18/`

Example 6.18 — listing

```
"""Calculate variance from Example 6.18."""

from __future__ import annotations

from statistics import fmean, pvariance, variance


def main() -> None:
    """Print deviations and both population and sample variance."""
    values: list[int] = [2, 4, 4, 6, 8]
    mean_value: float = fmean(values)
    squared_deviations: list[float] = [
        (value - mean_value) ** 2 for value in values
    ]
    print("Values:", values)
    print("Mean:", mean_value)
# ...
```

Range

- Range is max minus min, Example 6.20 on a small sample
- Simple but sensitive to outliers
- Pair range with quartiles or IQR for robust spread views in later plots

Measures of shape

- Skewness indicates asymmetry: right-skewed income tails pull the mean above the median
- Kurtosis relates to tail heaviness
- Example 6.2 from the introduction linked skewness to model choice

Frequency distributions and grouping

- Frequency tables count category occurrences or bin counts for numerics
- Group-by summaries compute mean spend by region or category
- Aggregation reveals segment differences raw global means hide

Takeaways

- Report center and spread together
- Shape guides transforms and model families
- Frequency and group-by tables bridge summaries to the visual techniques in the next parts

Next

- Complete the quiz for this part
- Continue to univariate and bivariate visualization